

SMART INDIA HACKATHON 2026



- **Problem Statement ID:** SIH26006
- **Problem Statement Title:** Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India
- **Theme:** Transportation & Logistics
- **PS Category:** Software
- **Team ID:**
- **Team Name:** Code Hackerz



An intelligent freight forecasting and chartering decision desk for SAIL's bulk imports into East Coast India.

PROPOSED SOLUTION: SYSTEM ARCHITECTURE

DETAILED EXPLANATION

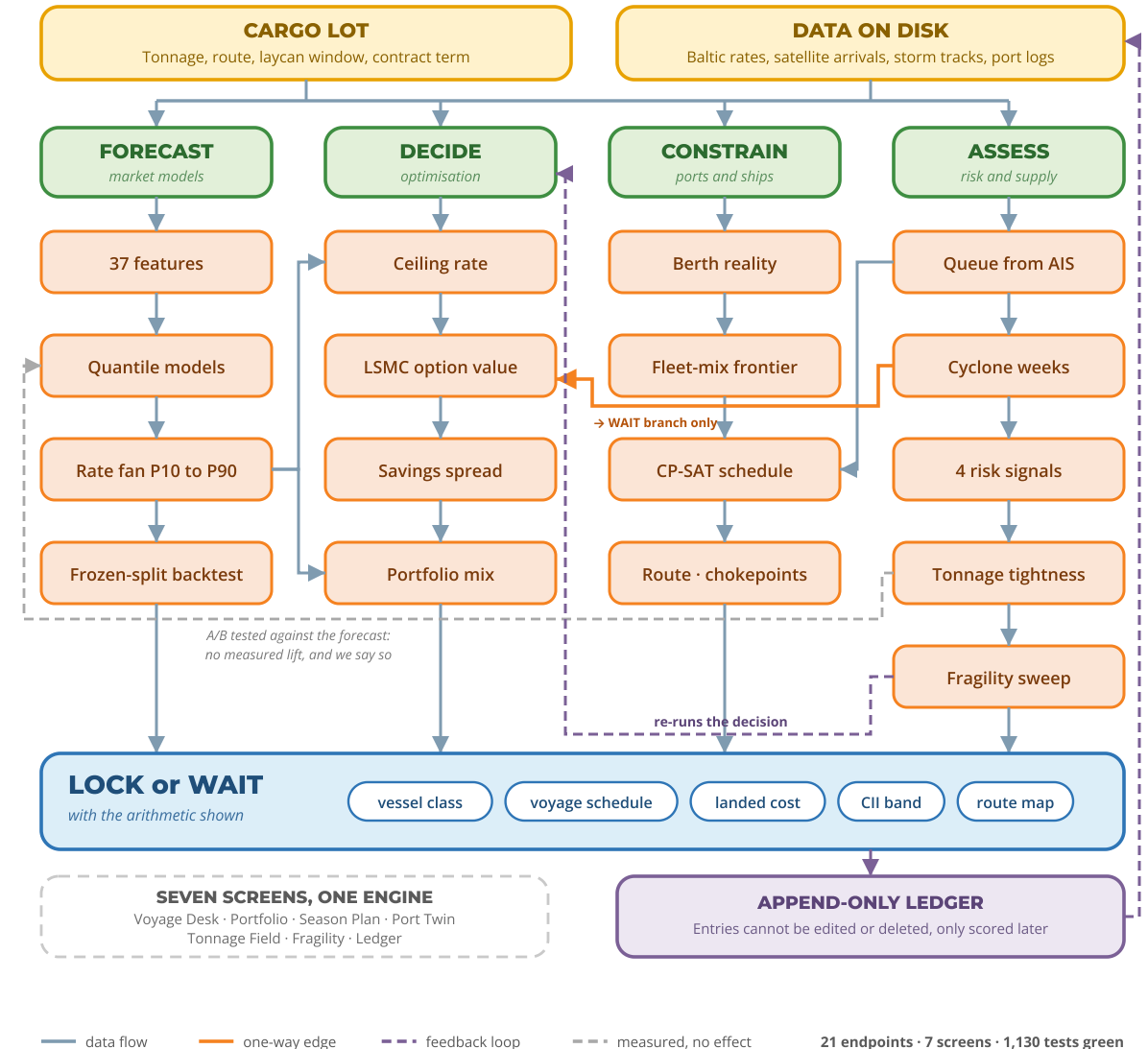
- **Forecast.** Quantile models over **136,933 rows** of real Baltic history return a low / mid / high rate band per vessel class at 7, 30 and 90 days.
- **Decide.** A ceiling rate caps what is worth paying today; a Longstaff-Schwartz solver prices the right to keep waiting, and expected weather delay is charged to the WAIT branch in dollars.
- **Constrain, then plan.** Every class is tested on draft, LOA, beam and DWT at **both** ports; a portfolio screen then splits the season across spot / period TC / CoA.

HOW IT ADDRESSES THE PROBLEM

- **(a) Market entry timing** → a dated LOCK/WAIT, with the trough window clamped inside the laycan.
- **(b) Vessel type optimization** → the cheapest **feasible** class off a full frontier, not a rule of thumb.
- **(c) Idle-time management** → queue time from satellite arrivals, CP-SAT sequencing, backhaul scored in dollars.
- **(d) Risk mitigation** → four independent live signals: rate regime, port congestion, chokepoint transits, cyclone climatology.

INNOVATION AND UNIQUENESS

- **It prices the option to wait**, worth **\$1,538/day** on a real run, moving the walk-away rate from \$21,016 to \$19,478.
- **It measures decisions, not accuracy:** realised saving against always-lock and always-wait, replayed on a frozen test split.
- **It refuses to invent numbers, and knows how wrong it can be.** A build check blocks fabricated data, and the fragility sweep found **3 cm** of draft margin.



One rule holds at every stage: never fabricate a number. Verified today by 1,130 passing tests.

TECHNOLOGIES USED

DATA & BACKEND

Python 3.12 FastAPI Polars
Pydantic v2 Parquet

MACHINE LEARNING

XGBoost quantile LightGBM
PyTorch LSTM pinball loss

OPTIMIZATION

OR-Tools CP-SAT LSMC
Ornstein-Uhlenbeck MC grid frontier

GEOSPATIAL & DOMAIN

searoute / Eurostat AIS port calls
IMO CII IBTrACS

INTERFACE

React + TypeScript Vite
hand-rolled SVG charts SSE streaming

METHODOLOGY AND PROCESS FOR IMPLEMENTATION

HOUSE RULES, ENFORCED AT EVERY STAGE

Never fabricate a number: a build-failing check scans the whole interface • A provenance label on every field • A named error, never a silent fallback • Deterministic: the same request returns identical figures

01 • HARVEST

-  **Baltic Exchange**
136,933 rows · 79 series
-  **IMF PortWatch (AIS)**
161 ports · 28 chokepoints
-  **NOAA IBTrACS v04r01**
726,241 storm fixes
-  **Port authority reports**
1,211 rows to 334 real calls
-  **Berth registers**
58 berths across 3 ports
-  **World Bank Pink Sheet**
ore, coal and FX
-  **IMO MEPC resolutions**
CII constants, primary PDFs
-  **GDELT 2.0**
1,484 weekly conflict rows
-  **Open-Meteo marine †**
7-day waves, cached 12 h

clean parquet on disk

02 • LEARN

FEATURE BUILD

37 features: lags to 63 d, 30/90-day rolling mean, std and z, returns, calendar, AIS calls, cross-index.

EMBARGOED SPLITS

9,512 train · 1,413 val · 818 test.
Guarded: touched once, at the end.

QUANTILE MODEL

XGBoost, pinball objective. P10 / P50 / P90 at 7/30/90 d × 4 classes.

MODEL GATE

Beat random walk, AR(1), LightGBM (tuned + untuned) and an LSTM.

MEASURED, FROZEN TEST SPLIT

HZN	PINBALL	RW	HIT
7 d	0.0357	0.0391	68.2%
30 d	0.0849	0.0925	63.0%
90 d	0.0886	0.1181	89.7%

rate fan · P10 / P50 / P90

03 • DECIDE

CEILING RATE

Highest daily rate at which locking beats spot, blended over the term across 7 / 30 / 90-day views.

OPTIMAL STOPPING · LSMC

Longstaff-Schwartz least-squares Monte Carlo returns an exercise boundary: a lock/wait price for every day of the horizon, not one threshold.

4,000 simulated paths

WEATHER TAX

Expected delay days × today's rate, added to the WAIT branch only. One-directional by construction.

SAVINGS DISTRIBUTION

2,000 Ornstein-Uhlenbeck paths give P10 / P50 / P90 saving and P(locking wins), seeded per request.

VERDICT

LOCK

WAIT

plus the boundary and the exact move that flips it.

boundary + verdict

04 • CONSTRAIN & WATCH

PORT FIT, BOTH ENDS

Draft · LOA · beam · DWT at both ends; rejected classes carry the binding reason.

FLEET-MIX FRONTIER

One configuration per class priced on the real forecast, cheapest feasible first. The verdict's class is taken from here.

CP-SAT SCHEDULER

OR-Tools vessel-to-parcel assignment with sequencing, real bunker price, demurrage.

ROUTING & CHOKEPOINTS

Eurostat sea-route graph, fallback flagged. Point-to-segment test on 28 chokepoints.

QUEUE + FOUR RISK CHANNELS

AIS arrivals vs the port's own 60-day norm, clamped to the priced date. z-scores on rates, port calls, strait transits + cyclone weeks.

feasible set + alerts

05 • SERVE & LEARN

COMMERCIAL LAYER

Landed cost as five labelled components, never summed blind. Backhaul + CII rating.

API + LIVE STREAM

FastAPI, 21 endpoints. /quote/stream emits one event per pipeline stage.

1,130 tests passing

TRADING-DESK UI

React + TypeScript, hand-rolled SVG charts. Every panel states its gaps.

APPEND-ONLY LEDGER

Written the moment it is made; per-entry edit and delete always raise. Scored against always-lock and always-wait.

FRAGILITY SWEEP

Re-runs the whole decision, binary-searching each input's flip point.

realised outcomes feed the decision-value backtest: five strategies replayed over the frozen split against a perfect-foresight oracle, evidence for the next retrain

† Open-Meteo is the one rolling feed: cached 12 h, degrades to no-signal and can never raise into a quote. Everything else is harvested at build time, to disk, and runs offline.

Not a mockup. 1,130 tests green, real public data, zero paid feeds, deployable today.

FEASIBILITY: A WORKING PROTOTYPE, MEASURED, NOT A PROPOSAL

1,130

Automated tests passing
109 files · full run 10m44s

21

Live HTTP endpoints
+ streamed quote pipeline

136,933

Rows of real market history
79 series · 2005 → 2026

16

Ports priced end to end
7 East Coast India · 9 load

4.5s

Tonnage rebuild, 128 ports
full sweep in seconds

30+

Self-audited faults closed
each re-verified

POTENTIAL CHALLENGES AND RISKS → STRATEGIES FOR OVERCOMING THEM

RISK

WHAT WE DID, AND WHAT REMAINS

Rate data is class-level, not route-level

Only 3 route-level observations exist for East India, below our own evidence bar. Distance, port fit, queue, fuel and landed cost *are* route-specific; the UI carries a badge on every quote. Closing it needs licensed data, not code.

Live feeds fail or are unavailable offline

Every external fetch is a build-time harvester writing to disk. The single rolling feed (marine forecast) is cached 12 h and degrades to no-signal, so it can never raise into a quote.

Port limits are not all published

Real berth registers for 3 ports (58 berths) are used where they exist; elsewhere the engine returns **CANNOT_VERIFY** and lowers its own confidence rather than guessing.

A model that looks confident but is wrong

Frozen, embargoed test split touched once. Fragility sweep reports the distance to a flipped answer. Append-only ledger makes cherry-picking structurally impossible.

One cargo lot priced at a time

Multi-parcel campaign optimisation is a real scope gap, deliberately deferred rather than half-built on the busiest path in the system, and it is listed as a gap, not omitted.

Adoption by a desk that trusts the phone

Every number ships with its provenance label and a written reason. A maintained fault register lists what is still wrong with the system, by name.

WHY IT IS VIABLE TO DEPLOY

- ▶ **Runs on a laptop.** Python + Node, one command each; no cluster, no GPU, no paid feed required to operate.
- ▶ **Deterministic by construction,** so the same request returns identical figures and a fixture can be reproduced and defended months later.
- ▶ **Data is public and free:** Baltic indices, IMF PortWatch AIS, NOAA IBTrACS, World Bank, IMO resolutions, port authority reports.
- ▶ **Extends beyond SAIL:** any bulk importer, any port set, since the port register and route graph are data, not code.

HONESTY AS AN ENGINEERING CONTROL

An automated check fails the build if anything resembling fabricated data appears in the interface.

Random values, hash-derived numbers and synthesised series are all rejected. Two engines that are built and tested but not yet wired to a screen are listed as exactly that, and **not** claimed as shipping features.

Judges can be handed the list of things that are wrong with this system. **Thirty-plus faults closed, each re-verified against its original evidence;** every remaining one carries a written reason.

IMPACT AND BENEFITS

On one real route it turned \$21,016 into \$19,478, and caught a 3 cm margin no other tool would.

POTENTIAL IMPACT ON THE TARGET AUDIENCE

- **SAIL chartering & logistics desk**, the primary user. Replaces the morning market call with a dated, defensible position.
- **Ministry of Steel / PSU importers**, who gain a common, auditable method across buyers.
- **Port & terminal operators**, given arrival and queue expectations stated in advance.
- **Plant & production planning**, with coverage tied to real days of buffer stock, not a preference slider.

Designed to be argued with, not obeyed. Every verdict expands into the real factors that produced it, so a chartering manager can overrule it for a reason the model cannot see, and that reason is recorded.

ECONOMIC

- **A walk-away price on every deal.** The desk stops paying above a rate it can justify, and knows the exact rate move that flips the call.
- **Spot to period shift, quantified.** The efficient frontier prices what buying certainty actually costs across the season.
- **Measured on realised saving** versus always-lock and always-wait, not on model accuracy.

OPERATIONAL

- **No infeasible fixture.** Draft, LOA, beam and DWT are checked at both ends before signing, not at the pilot station.
- **Less idle time.** Queue time from real satellite arrivals; CP-SAT sequencing; ballast legs scored against real cargo probability.
- **Fewer surprises.** Four disruption channels watched continuously per route.

ENVIRONMENTAL

- **Carbon rating before the fixture.** IMO attained-CII arithmetic projects the A to E band the voyage would earn the vessel, with constants read from the primary resolutions.
- **Less deadheading** and less idling at anchor is directly less fuel burnt and less CO₂ emitted.

SOCIAL & GOVERNANCE

- **Public-money decisions become auditable.** Every recommendation is written to an append-only record the moment it is made, and no entry can be edited or deleted.
- **Reliable steel supply.** Stockout risk is priced against real plant cover days.
- **Institutional knowledge outlives the individual** who made the call.

WORKED EXAMPLE, REAL RUN

**Newcastle → Paradip
75,000 t**

Every figure below came off the running system.

RIGHT SHIP, FOR A STATED REASON

2 × Supramax

Panamax **rejected** on Paradip's deadweight limit, and the verdict panel agrees with it.

THE OPTION TO WAIT, IN MONEY

\$1,538/day

Moved the walk-away rate from **\$21,016** to **\$19,478**, a reversal a naive comparison cannot produce.

HOW WRONG CAN WE BE?

3 cm

Permissible-draft margin before the recommendation flips. No accuracy metric surfaces that.

DATA SOURCES: ALL REAL, ALL ON DISK, ALL OFFLINE-SAFE

SOURCE	WHAT IT IS, AND WHAT IT FEEDS
Baltic Exchange balticexchange.com	BCI · BPI · BSI · BHSI · BDI and time-charter averages, 2005 → 2026-08-20. 136,933 rows, 79 series , giving the forecast target and today's real daily rate.
IMF PortWatch portwatch.imf.org	AIS-derived daily port calls at 161 ports and daily transits + coordinates for 28 chokepoints . Feeds congestion, queue time, tonnage supply, backhaul, chokepoint alerts.
NOAA IBTrACS v04r01 ncei.noaa.gov/products/ibtracs	726,241 storm fixes (331 MB), reduced to a per-basin, per-ISO-week strike climatology. Feeds the cyclone alert and the weather delay.
IMO MEPC resolutions 308(73) · 338(76) · 352(78) · 353(78) · 354(78)	CII reference lines, fuel-to-CO ₂ factors, reduction factors and A to E boundaries, read from the primary PDFs and checked against the resolution's own worked example.
Eurostat marine network via searoute	Real sea-route graph for distance and routing; great-circle fallback is flagged in the response, never silent.
World Bank Pink Sheet	Iron ore and coal benchmark prices and FX, the landed-cost commodity component.
Port authority daily traffic reports	1,211 report rows → 334 distinct real calls ; plus 58 declared berth rows across Visakhapatnam, Dhamra and Gangavaram.
Open-Meteo Marine API	Seven-day wave-height forecast at both ports, the only rolling feed in the system. Cached 12 h, degrades to no-signal offline, documented as a deliberate exception.
Signal Ocean route assessments	Real route-level rate quotes. Reported here because they produced an <i>honest negative</i> : only one series is a genuine East-Coast-India rate, on three observations, below our evidence bar, so it is not used.
GDELT 2.0 · Joint War Committee	12 months of conflict-event counts across 28 chokepoints and current war-risk Listed Areas, feeding the fracture index (built, not yet exposed).

METHODS AND LITERATURE

Longstaff & Schwartz (2001)*Rev. Financial Studies 14(1)*

Least-squares Monte Carlo for American options: how we price the right to keep waiting, as an exercise boundary per day.

Almgren & Chriss (2000)*Journal of Risk 3(2)*

Optimal execution of large orders, transplanted to freight so SAIL's own buying does not move the rate against itself.

Chen & Guestrin (2016), XGBoost

Gradient-boosted trees with a pinball (quantile) objective at P10 / P50 / P90.

Uhlenbeck & Ornstein (1930)

Mean-reverting price paths for the savings distribution. Freight rates revert, and a flat random walk overstates long-contract risk.

Google OR-Tools CP-SAT

Vessel↔cargo assignment with sequencing under hard port-feasibility constraints.

Koenker & Bassett (1978)

Quantile regression and the pinball loss we score every model on.

PROJECT DOCUMENTATION, SHIPPED WITH THE CODE

- **Technical reference:** every engine, its method, and the reason that method was chosen over the alternatives.
- **Architecture note:** the components, the data flow, and the responsibility boundary each layer is held to.
- **Model guide:** how to add a new forecasting model and compare it fairly against what already exists.

Provenance is a first-class field, not a footnote.

Five labels (**OBSERVED · ESTIMATED · INFERRED · MODEL_DERIVED · DECLARED**) are applied per field, and a value computed from a real input is at best MODEL_DERIVED. It never inherits its input's label.